

Inconel 600 | High-Temperature & Corrosion-Resistant Nickel-Chromium Alloy

Introduction:

Inconel 600 (UNS N06600) is a nickel-chromium-iron alloy renowned for high-temperature strength, excellent oxidation, and corrosion resistance. It retains mechanical properties in both oxidizing and reducing environments and is widely used in chemical, marine, aerospace, power generation, and industrial furnace applications.

Chemical Composition:

Element	Composition (%)
Nickel	72.0 min (Balance)
Chromium	14.0 - 17.0
Iron	6.0 - 10.0
Carbon	0.15 max
Manganese	1.0 max
Sulfur	0.015 max
Silicon	0.5 max
Copper	0.5 max

Physical & Mechanical Properties:

- Density: 8.47 g/cm³
- Melting Range: 1354-1413 °C
- Tensile Strength: 550-1520 MPa
- Yield Strength: 170-1450 MPa
- Elongation: 2-55%
- Hardness: 24C-90B

Available Products:

- Seamless & Welded Pipes (ERW & EFW)
- Tubes (Capillary, U-Bend)
- Sheets, Plates, Coils & Strips
- Round, Flat, Hex & Square Bars
- Pipe Fittings, Flanges & Fasteners
- Custom Fabricated Components

Applications:

- Heat Treatment & Industrial Furnaces (retorts, trays, radiant tubes)
- Chemical Processing (vessels, piping, heat exchangers)
- Nuclear & Power Generation (reactor components, steam generators)
- Food Processing & Pharmaceuticals (acidic systems, hygienic equipment)
- Aerospace & Jet Engines (engine parts, afterburners, exhaust liners)
- Marine & Offshore Systems (pumps, heat exchangers, risers)
- Cryogenic Applications (sub-zero storage and transfer systems)

Advantages:

- Exceptional oxidation and corrosion resistance at high temperatures
- Resistant to chloride-induced stress corrosion cracking
- Retains strength in oxidizing and reducing environments
- Good weldability and machinability
- Reliable performance in extreme thermal and chemical conditions

Why Choose Moksh Tubes & Fittings LLP:

- Specialists in high-performance nickel alloys
- ISO-certified supply with global logistics support
- Custom fabrication and machining capabilities
- Trusted by chemical, marine, aerospace, and industrial clients